

Initial Product Backlog

AI-Driven Robotic Manipulation: Cross-Embodiment Learning with Panoptic Scene Graphs

Team Name: AI-Driven Robotic Manipulation: Integrating Learning Models with Real-World Control

Team Members: Toan Pham, Albert Vo

Product Owner/Sponsor: Dr. Nakul Gopalan

Date: Sep 23, 2025

Executive Summary

This initial product backlog outlines the development of a morphology-agnostic robot control system that enables policy transfer between Franka Emika and Sawyer robots using panoptic scene graphs as intermediate representations. The project leverages state-of-the-art vision-language models, scene understanding algorithms, and cross-embodiment learning techniques to achieve zero-shot transfer capabilities.

Customer Archetypes

Primary Customers

Capstone Sponsor / Faculty Advisor (Dr. Nakul Gopalan)

Needs: Working prototype, well-documented results, validated cross-embodiment transfer

Pain Points: Long training times, sim-to-real gap, lack of generalizable solutions

Capstone Users and Sponsor Team

Needs: Reusable frameworks, documented code, reproducible experiments

Pain Points: Platform-specific solutions, expensive retraining for new robots

Industry Partners (Future Users)

Needs: Flexible manipulation systems, reduced deployment costs, multi-robot coordination

Pain Points: Robot-specific programming, lack of skill transfer

EPICs

EPIC-1: Environment and Infrastructure Setup

Priority: Critical

Estimated Duration: 2 sprints

Description: Establish complete development environment including RoboSuite simulation, Sol supercomputer access, and baseline data collection pipeline.

EPIC-2: Vision-Language Foundation

Priority: High

Estimated Duration: 2 sprints

Description: Integrate Segment Anything Model (SAM) and language-guided segmentation for robust object detection and scene understanding.

EPIC-3: Panoptic Scene Graph Generation

Priority: Critical

Estimated Duration: 3 sprints

Description: Implement and optimize scene graph generation using VLPrompt/DSFormer for creating morphology-agnostic representations.

EPIC-4: Cross-Embodiment Policy Learning

Priority: Critical

Estimated Duration: 3 sprints

Description: Develop transformer-based policy architecture supporting both Franka and Sawyer embodiments with scene graph conditioning.

EPIC-5: Transfer Learning and Validation

Priority: High

Estimated Duration: 2 sprints

Description: Achieve and validate zero-shot transfer between robot platforms with comprehensive testing and ablation studies.

Initial User Stories (Prioritized)

US-001: RoboSuite Environment Configuration

EPIC: EPIC-1

Priority: P0 (Must Have)

Story Points: 8

As a researcher, I want to configure RoboSuite with Franka and Sawyer models, so that I can simulate both robot platforms for data collection.

Acceptance Criteria:

RoboSuite installed with MuJoCo dependencies

Franka and Sawyer models loading correctly

Basic teleoperation working for both robots
10+ test trajectories collected per robot
Documentation of setup process

US-002: Sol Supercomputer Setup

EPIC: EPIC-1

Priority: P0 (Must Have)

Story Points: 5

As a researcher, I want to access Sol supercomputer resources, so that I can train large-scale models efficiently.

Acceptance Criteria:

Sol account activated with GPU access

SLURM job submission working

PyTorch environment configured with CUDA

Test job achieving >90% GPU utilization

Storage directories organized (home/scratch/project)

US-003: SAM Integration

EPIC: EPIC-2

Priority: P1 (Should Have)

Story Points: 13

As a system, I want to segment objects using SAM, so that I can provide pixel-accurate masks for manipulation.

Acceptance Criteria:

SAM model loaded (ViT-B or ViT-H variant)

Real-time inference (<100ms per frame)

Integration with RoboSuite camera feeds

90%+ IoU on standard objects

Support for language-guided segmentation

US-004: Scene Graph Generator Implementation

EPIC: EPIC-3

Priority: P0 (Must Have)

Story Points: 21

As a system, I want to generate panoptic scene graphs from visual observations, so that I can create morphology-agnostic task representations.

Acceptance Criteria:

VLPrompt or DSFormer implemented

Scene graphs generated at 5Hz minimum

Object relations correctly identified (80%+ accuracy)

Integration with policy input pipeline

Visualization tools for debugging

US-005: Cross-Embodiment Dataset Collection

EPIC: EPIC-4

Priority: P1 (Should Have)

Story Points: 13

As a researcher, I want to collect demonstrations on both robots, so that I can train cross-embodiment policies.

Acceptance Criteria:

8+ demonstrations per robot

5+ different manipulation tasks

Consistent task definitions across robots

Data format supporting both embodiments

Quality metrics (success rate >70%)

US-006: Transformer Policy Architecture

EPIC: EPIC-4

Priority: P0 (Must Have)

Story Points: 21

As a system, I want to learn policies conditioned on scene graphs, so that I can execute tasks independent of robot morphology.

Acceptance Criteria:

Transformer architecture implemented

Scene graph conditioning working

Separate action heads per robot

Training convergence in <48 hours

Baseline performance >50% success rate

US-007: Zero-Shot Transfer Validation

EPIC: EPIC-5

Priority: P0 (Must Have)

Story Points: 13

As a researcher, I want to demonstrate zero-shot transfer, so that I can validate morphology-agnostic learning.

Acceptance Criteria:

Franka policy tested on Sawyer tasks

Sawyer policy tested on Franka tasks

Transfer success rate >40%

Comparison with baseline methods

Statistical significance demonstrated

US-008: Ablation Studies

EPIC: EPIC-5

Priority: P2 (Nice to Have)

Story Points: 8

As a researcher, I want to conduct ablation studies, so that I can identify critical components for transfer.

Acceptance Criteria:

Test with/without scene graphs

Compare different graph architectures

Analyze failure modes

Quantify component contributions

Risk Management

Technical Risks

Risk 1: Simulation-to-Real Gap

Impact: High - May invalidate transfer results

Probability: Medium

Mitigation: Domain randomization, early real robot testing if possible

Owner: Both team members

Risk 2: Scene Graph Quality

Impact: High - Poor graphs prevent transfer

Probability: Medium

Mitigation: Multiple graph generation methods, manual annotation backup

Owner: Albert Vo

Risk 3: Training Non-Convergence

Impact: Medium - Delays progress

Probability: High

Mitigation: Start with proven baselines, extensive hyperparameter search

Owner: Toan Pham

Resource Risks

Risk 4: Sol GPU Availability

Impact: High - Blocks training

Probability: Low

Mitigation: Off-peak scheduling, backup cloud resources

Owner: Toan Pham

Risk 5: Data Collection Bottleneck

Impact: Medium - Limits diversity

Probability: Medium

Mitigation: Automated collection, data augmentation

Owner: Albert Vo

Timeline Risks

Risk 6: Integration Complexity

Impact: Medium - Causes delays

Probability: High

Mitigation: Modular architecture, early integration tests

Owner: Both team members

Technical Debt Considerations

Framework Dependencies: Lock versions early to avoid breaking changes

Data Format Standardization: Define consistent format across robots from start

Checkpoint Management: Implement robust saving/loading from Sprint 1

Documentation: Maintain as code develops, not after

Definition of Done (Global)

☐ Code reviewed and merged to main branch

☐ Unit tests passing (where applicable)

☐ Integration tests passing

☐ Documentation updated (README, docstrings)

☐ Experimental results logged in lab notebook

☐ Metrics tracked in Weights & Biases

☐ Sprint demo prepared

☐ Acceptance criteria verified

Meeting Notes with Product Owner

Date: Sep 23, 2025

Attendees: Dr. Nakul Gopalan, Toan Pham, Albert Vo

Discussion Points:

Reviewed related cross-embodiment learning literature for zero-shot transfer insights

Discussed intermediate representation approaches from RSS 2020 work

Agreed on panoptic scene graphs as primary abstraction

Set expectations: Simulation-only for first 8 weeks, real robots if time permits

Key Decisions:

Focus on Franka-Sawyer transfer (most different morphologies available)

Use RoboSuite for consistency with literature

Leverage existing VLPrompt/DSFormer rather than custom scene graphs

Weekly progress meetings on Wednesdays

Product Owner Agreement

I have reviewed the Initial Product Backlog and agree with the priorities, scope, and approach outlined for the cross-embodiment robot learning project.

Signature:

A handwritten signature in red ink, consisting of a large, stylized 'N' followed by a horizontal line and a small loop.

Dr. Nakul Gopalan
Date: Sep 23, 2025